

INNOVA JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION 2
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

CLASS

INDEX NUMBER

BIOLOGY

9648/03

Paper 3 Applications Paper

23 September 2014

2 hours

Additional Materials: Answer Paper
Cover Page

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, fasten all your work securely together.

The number of marks is given in the brackets [] at the end of each question or part question.

For Examiner's Use	
1	15
2	13
3	12
4	12
5	20
Total	72

This document consists of **18** printed pages.

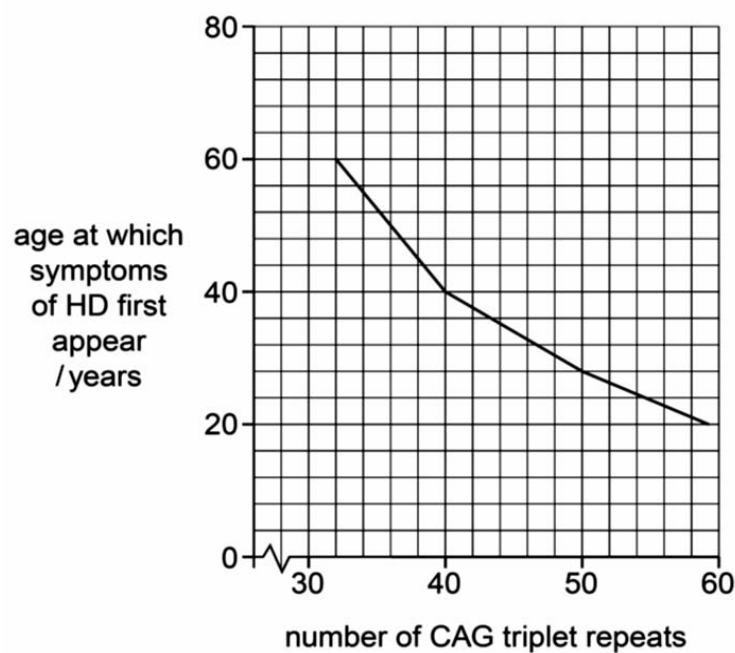


Section AAnswer **all** questions.

- 1** Huntington's disease (HD) is an autosomal dominant disorder characterised by motor dysfunction, cognitive decline and psychiatric disturbances, which lead to progressive dementia and death approximately 15 – 20 years after disease onset.

HD is caused by CAG repeats in the *IT15* gene coding for the huntingtin (HTT) protein. Increase in the number of CAG repeats coding for a glutamine-rich domain in HTT protein seems to induce conformational changes in the protein which causes it to aggregate within cells leading to cell death.

The age at which symptoms of HD first appear is linked with the number of CAG repeats. This is shown in Fig. 1.1.

**Fig. 1.1**

- (a)** Describe the relationship between the number CAG repeats and the age at which symptoms of HD first appear.

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..... [2]

Transgenic mice expressing the dominant HD allele develop many of the symptoms of human HD. The progress of the disease in the transgenic mice was monitored by measuring how long the mice could maintain their grip on a rotating rod up to a maximum of 10 minutes (Fig. 1.2). The rod was rotated at 5, 15 or 25 revolutions per minute (rpm).



Fig. 1.2

The performance on the rotating rod of 9- and 12-week old mice from transgenic HD mice was compared to normal control mice. The results are shown in Fig. 1.3.

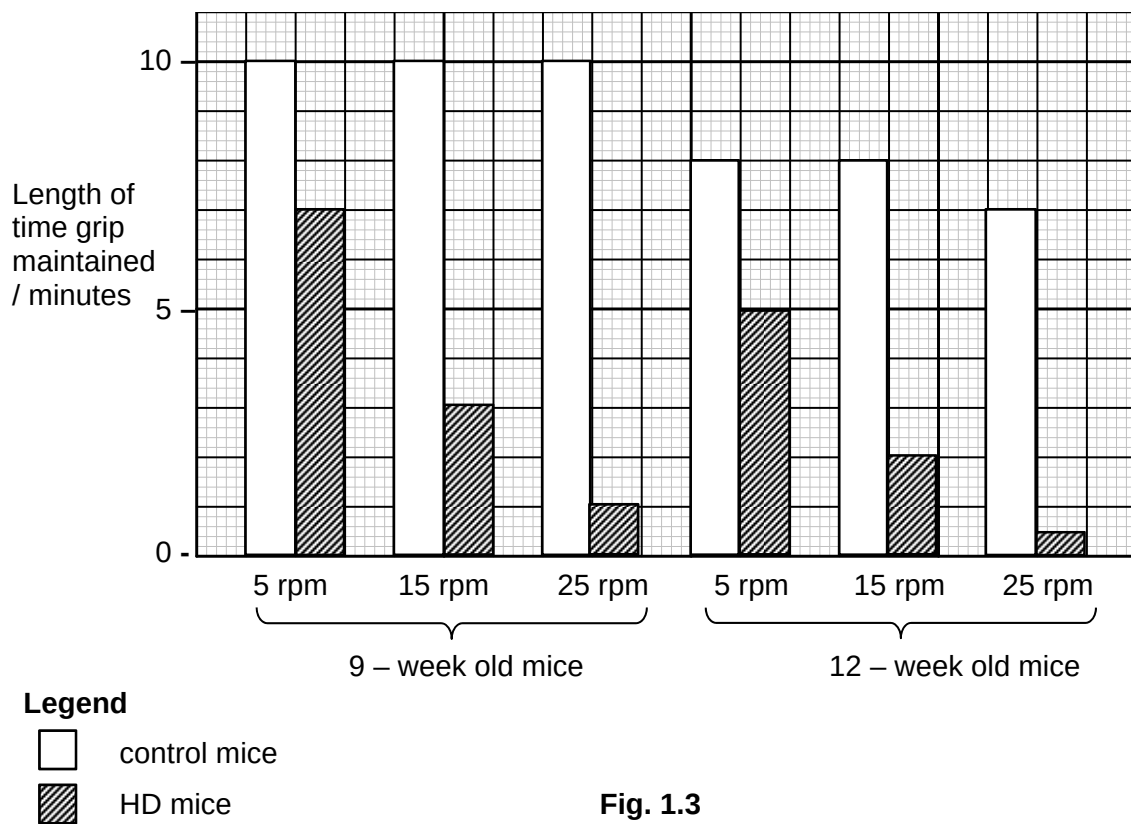


Fig. 1.3

- [2]

- [2]

- (i) Explain how restriction fragment length polymorphism, agarose gel electrophoresis and nucleic acid hybridisation can be used in the blood test to estimate the age at which HD symptoms first appear for individuals.

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- (ii) The location of bands obtained from (c)(i) for different individuals were found to be very close to each other. This made it difficult to determine the size of the DNA fragment. Suggest **one** change to the procedure that can improve the process.

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..... [1]

- (d) Explain if HD is likely to be effectively treated using gene therapy.

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..... [2]

[Total: 15]

- 2 Severe Combined Immunodeficiency (SCID) is a group of genetic disorders where patients display severe immunodeficiency and are therefore extremely vulnerable to infectious diseases.

Fig. 2.1 shows the molecular basis of a particular form of SCID where the IL-R2 γ gene is mutated.

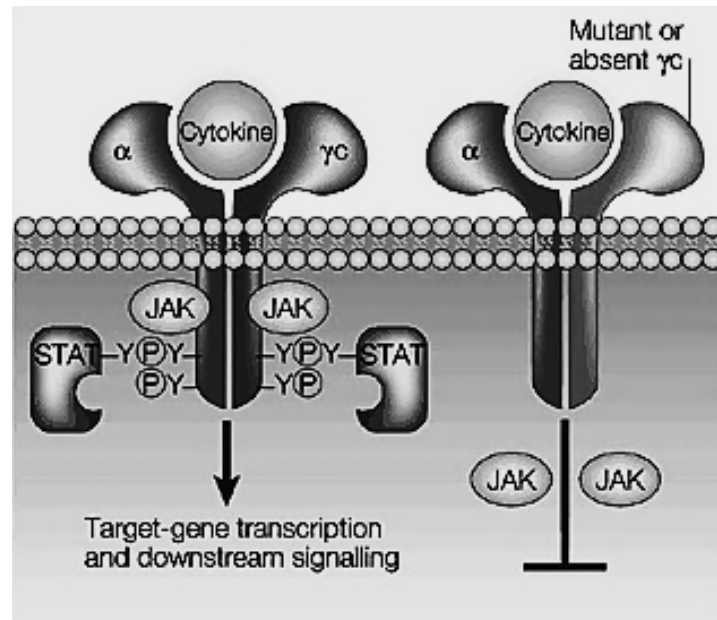


Fig. 2.1

(a) With reference to Fig. 2.1,

- (i) state the chromosome on which the IL-R2 γ gene is found;

[1]

- (ii) describe the molecular basis of the SCID shown.

[3]

- (b) Early treatment of SCID involved the genetic modification of the patient's own lymphocytes. More recent treatments used genetic modification of the patient's own stem cells. In both cases the viral gene delivery system was used.

- (i) Describe **two** problems that may arise when using a viral vector to introduce an allele into a cell.

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..... [2]

- (ii) Explain why the more recent gene therapy has proved more effective.

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..... [2]

- (c) In 2006, scientists in Japan showed that the introduction of four specific genes could reprogramme mature adult cells into a type of pluripotent stem cell called the induced pluripotent stem cell (iPSC). This potentially can replace the usage of embryonic stem cells in gene therapy.

- (i) Describe the characteristics of a stem cell.

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..... [3]

- (ii) State **one** argument in favour of the use of iPSCs.

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..... [1]

- (iii) Recently, iPSCs were discovered to retain physiological memory of the adult cell it was derived from. This includes the inability of the cells to synthesise particular proteins coded by its genome.

Suggest why this is so.

----- [1]

[Total: 13]

- 3 To boost food production for a rapidly growing global population, crop yields must significantly increase. The C₄ Rice Project aims to improve the photosynthetic capacity of C₃ crops like rice.

Fig. 3.1 shows the production of transgenic C₄ rice plants by the *Agrobacterium*-mediated transformation of rice using calli induced from mature seeds. The genes involved in the C₄ photosynthetic pathways, antibiotic resistance gene and the genes coding for green fluorescent protein (GFP) were added to form the recombinant plasmid in the bacteria.

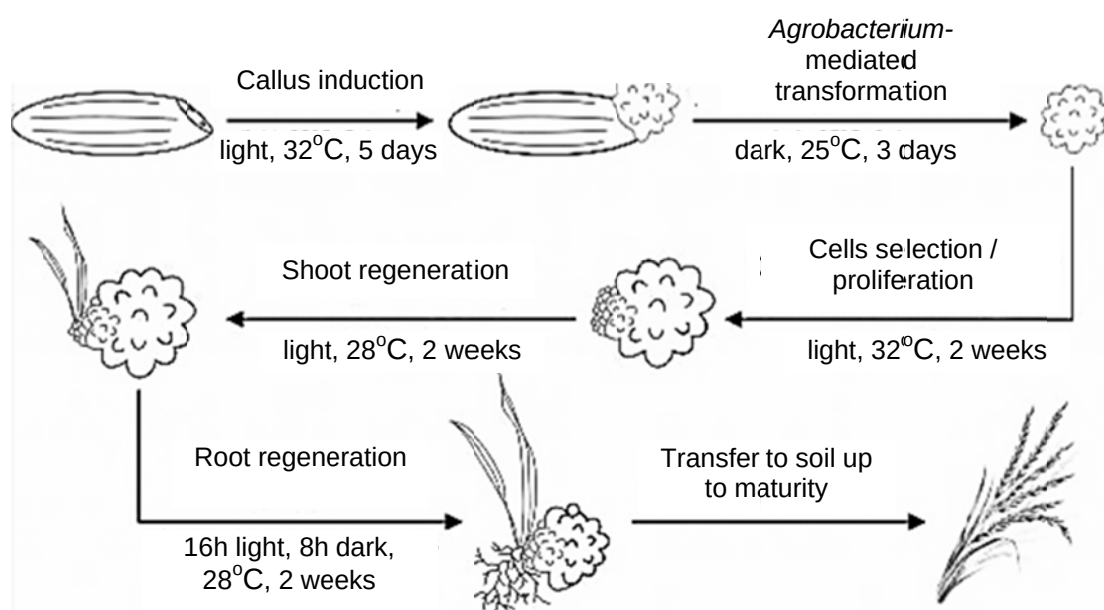


Fig. 3.1

(a) With reference to Fig. 3.1,

- (i) describe the process of callus induction.

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..... [3]

- (ii) explain the significance of the GFP in the selection of transformed tissues.

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..... [1]

- (b) Suggest **one** advantage of using seeds as explants.

[1]

Research suggests that C_4 rice is able to increase the efficiency of utilizing solar radiation that can be translated into yield. Fig. 3.2 shows the net photosynthetic rate of C_3 and C_4 leaves at different light intensity.

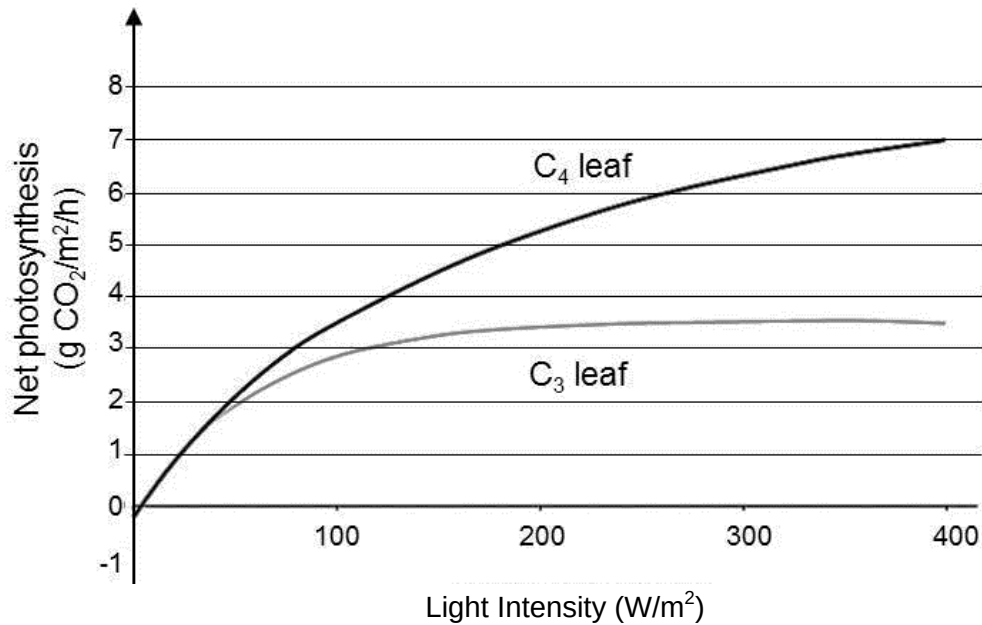


Fig. 3.2

- (c) With reference to Fig. 3.2, describe the difference in net photosynthetic rate in the C_3 and C_4 leaves.

[3]

One of the main contributing factor to a higher photosynthetic efficiency in C_4 plants is the absence of a process known as photorespiration. Photorespiration occurs when partial pressure of oxygen is high relative to carbon dioxide. It results in lower carbon dioxide fixation in C_3 plants. This is due to the enzyme Rubisco catalysing a reaction between oxygen and RuBP that results in the production of non-useful by-products. During this process, ATP and reduced NADP are used without leading to the production of sugar.

In C_4 plants, photorespiration is avoided by special arrangement of cells involved in photosynthesis. Additional reactions occur during the light independent stage in mesophyll cells while the Calvin cycle occurs in separate cells known as bundle sheath cells. Fig. 3.3 shows the arrangement of these cells that allows carbon dioxide to be concentrated in the bundle sheath cell.

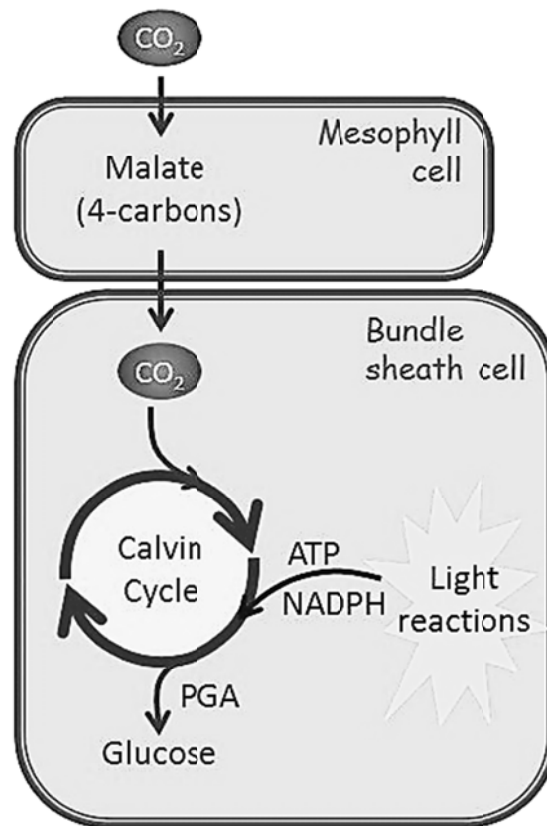


Fig. 3.3

- (d) With reference to Fig. 3.3 and information provided, suggest why C_4 plants are more effective at carbon dioxide fixation at high light intensity.

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..... [3]

- (e) Suggest **one** concern of genetically modifying a C_3 plant to a C_4 plant with a higher photosynthetic efficiency.

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..... [1]

[Total: 12]

4 Planning question

Dormant seeds have a very low rate of respiration. When water is absorbed by dormant seeds, growth hormones are activated. These hormones activate genes that code for enzymes that hydrolyse stored food reserves used in respiration allowing the seed to grow and germinate. The respiration rate can be measured by oxygen usage per unit mass using a respirometer.

You are required to investigate the respiration rate of germinating seeds of different mass.

You are provided with the following materials and apparatus which you must use:

- mung bean seeds of different mass that have been soaked in water for 24 hours
- blue dye
- soda lime pellets
- glass beads
- rubber tubing connected to capillary tube of 4mm diameter
- syringes
- electronic weighing balance

You may select from the following apparatus and use appropriate additional apparatus:

- normal laboratory glassware e.g. test-tubes, beakers, measuring cylinders, graduated
- pipettes, glass rods etc.
- forceps
- ruler
- stopwatch
- thermometer

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagram(s), if necessary,
- identify the independent and dependent variables,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- show how you will record your results and the proposed layout of results tables and graphs,
- use the correct technical and scientific terms,
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 12]

This image shows a full page of a worksheet designed for handwriting practice. It consists of approximately 20 horizontal dashed lines spaced evenly across the page, providing a guide for letter height and placement. The background is plain white, and there are no other markings or text present.

This image shows a full page of a worksheet designed for handwriting practice. It features 20 evenly spaced, horizontal dashed lines across the entire page, providing a guide for letter height and placement. The background is plain white, and there are no margins or additional markings.

[illegible]

Free-response question

Write your answers to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labelled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections **(a)**, **(b)**, etc., as indicated in the question.

5 (a) Discuss the problems brought about by the human genome project. [6]

(b) Describe the advantages and disadvantages of plant cloning. [6]

(c) With reference to solving the demand for food in the world, explain, with an example, the significance of genetically engineered animals to improve their quantity. Include, in your explanation, one concern and one benefit of such GMO. [8]

[Total: 20]

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